Model-Free Control of Longitudinal Vehicle Dynamics Using Algebraic Derivative Estimation: Implementation and Analysis in MATLAB/Simulink

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***Abstract*— This paper presents the design, implementation, and evaluation of the Model-Free Control architecture for longitudinal vehicle speed regulation, using Algebraic Derivative Estimation. The objective is to track a prescribed reference velocity profile without requiring an explicit vehicle model, based on ultra-local modeling principles. ADE is used to estimate the longitudinal acceleration in real time, enabling robust feedback without numerical differentiation noise. The controller is implemented in MATLAB/Simulink with a validated fixed-step simulation environment. Additional subsystems, including torque-rate constraints and an Ego Vehicle dynamic model, are integrated to create a complete closed-loop system. Simulation results demonstrate effective tracking performance, noise rejection characteristics, the influence of ADE window size N, and controller behavior under varying torque-rate constraints.**

Keywords — Model-free control, Algebraic derivative estimation, ultra-local modeling, Ego vehicle dynamics.

I. Introduction

Speed control in the longitudinal direction forms the basis for the development of advanced driver assistance systems and autonomous vehicles. Traditional model-based control approaches include PID, adaptive control, and full-state feedback, which generally rely on vehicle parameters that are either difficult to measure accurately or vary with operating conditions, hence creating difficulties related to robustness, tuning, and deployment.

The Model-Free Control, introduced by Fliess and Join, is based on substituting detailed system models by online estimations directly obtained from sensor measurements. Instead of explicit modeling of complex tire-road interactions, resistive forces, drivetrain delays, or aerodynamic loads, MFC invokes the ultra-local model:

$$\dot{v}(t) = F(t) + \alpha u(t) \tag{1}$$

where $u(t)$is the input torque, $v(t)$is the longitudinal speed, $F(t)$ combines all unknown dynamics, and α is a chosen scalar representing input effectiveness. F(t) is estimated in real time using Algebraic Derivative Estimation (ADE), enabling high-performance control without parameter identification.

This project implements the full MFC architecture in MATLAB/Simulink for a longitudinal vehicle model. ADE is used to compute a filtered estimate of acceleration from noisy velocity measurements. The model also includes torque constraints and configurable torque-rate limits. The system performance is evaluated through a series of simulations, including different ADE window sizes.

Figures such as the Simulink top-level model (Fig. 1), ADE subsystem (Fig. 2), and MFC torque controller (Fig. 3) illustrate the implementation architecture and are referenced throughout the paper.

II. Background Theory

A. Longitudinal Vehicle Dynamics

The longitudinal dynamics of a ground vehicle can be approximated as:

$$m\dot{v}_x = T_r - F_{\text{res}} \tag{2}$$

where,

- v_x= longitudinal velocity,
- T_r= resultant traction torque at the wheels,
- F_{res} = resistive forces (rolling, aerodynamic drag, grade).

In traditional modeling, one must compute:

$$F_{\text{res}} = mgC_r + \frac{1}{2}\rho A_f C_d v_x^2 + mg\sin(\theta) \tag{3}$$

which requires knowledge of vehicle mass, drag coefficient, road slope etc. In contrast, MFC removes this requirement entirely by absorbing all unknowns into a single term, $F(t)$.

B. Ultra-Local Model

The ultra-local model used in MFC is defined as (1) where:

- α is selected by the designer (typically based on input magnitude scaling),
- $F(t)$ contains all unmodeled dynamics and disturbances,
- $u(t)$is the torque command.

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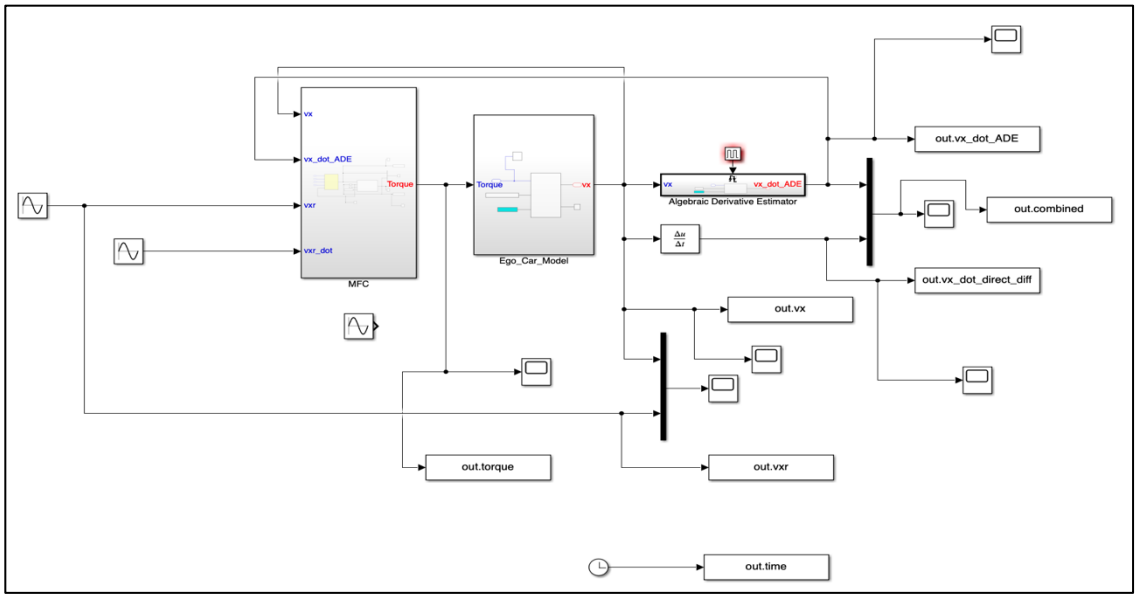

Fig 1 Simulink top-level model

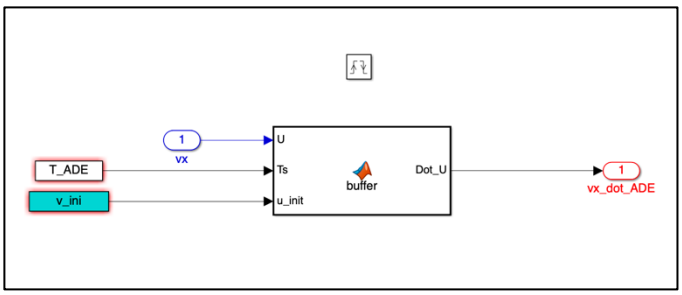

Fig 2 ADE subsystem

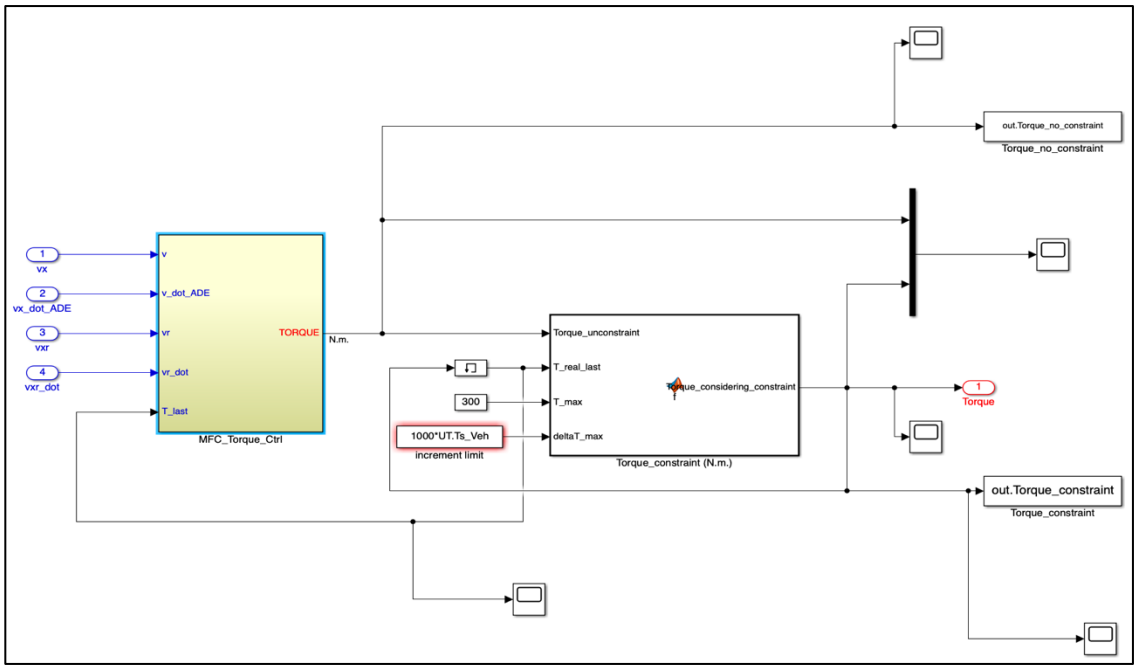

Fig 3 MFC torque controller

The model does not assume any physical structure. Instead, the unknown component $F(t)$ is estimated continuously from real-time measurements using ADE.

C. Algebraic Derivative Estimation (ADE)

ADE is used to estimate derivatives from noisy signals without numerical differentiation. The algorithm uses a sliding window of N past samples of $v(t)$ and applies integral-based polynomial reconstruction.

The general form for ADE derivative estimation is:

$$\hat{v}(t) = \sum_{k=0}^{N} P_k v(t - kT_s) \tag{5}$$

where:

- P_k= weights determined by the ADE formula,
- N= sliding window length,
- T_s= sampling period.

The ADE implementation in this project directly follows the formulation from Lecture 21 (Model-Free Control) and Lecture 7 (Vehicle Simulation Fundamentals).

Appendix B shows the ADE Simulink block using the MATLAB Function implementation.

D. Model-Free Controller (MFC)

Using the ultra-local model the general form for ADE derivative estimation is:

$$\hat{F} = \hat{v} - \alpha u(t - T_s) \tag{6}$$

The control law is:

$$u = \frac{-\hat{F} + \dot{v}_r + \eta(v - v_r)}{\alpha} \tag{7}$$

where:

- v_r= reference velocity,
- $\dot{v}_r$= reference derivative,
- η= tracking gain,
- $\hat{F}$= ADE-based estimate.

This results in a PI-like behavior with instantaneous feedforward compensation.

The complete MATLAB Function block Appendix A is implemented exactly from lecture notes.

E. Torque Constraint Logic

For realistic operation, electric motors and IC engines have limits on:

- maximum torque magnitude $T_{\max}$
- maximum torque rate $\Delta T_{\max}$

The project includes a dedicated Simulink subsystem enforcing:

$$|T| \leq T_{max} \tag{8}$$

$$|T(t) - T(t - T_s)| \leq \Delta T_{max} \tag{9}$$

III. SYSTEM ARCHITECTURE

The complete longitudinal velocity control system is implemented in MATLAB/Simulink, following a modular structure aligned with the MFC formulation. The top-level model is shown in Fig. 1, where all major subsystems—MFC controller, ADE estimator, Ego Vehicle dynamics, torque constraint logic, and reference generator—interact in a closed loop.

A. Simulink top level structure

This diagram (Fig. 1) shows:

- Inputs: reference velocity v_r, reference derivative $\dot{v}_r$
- ADE subsystem
- MFC subsystem
- Ego Vehicle Model
- Data logging blocks
- Output channels: v_x , $\dot{v}_x$_ADE , torque, and direct derivative

The system uses a fixed-step ODE1 solver with sampling time $T_s = 0.01\ s$.

B. Algebraic Derivative Estimation (ADE) Subsystem

The subsystem receives measured velocity v_x and outputs $\dot{v}_x$_ADE . The internal algorithm is implemented using a MATLAB Function block shown in Fig. 2.

This code corresponds to (5). The sliding window length N is configurable and later analyzed in Section V.

C. MFC Torque Controller

Inputs:

- v(measured speed)
- $\dot{v}$_ADE
- reference v_r, $\dot{v}_r$
- previous torque T_{last}

Output:

- Unconstrained torque command

The controller applies (6) and (7).

D. Torque Constraint subsystem

This block enforces:

- Maximum torque $T_{\max} = 300$ Nm
- $\left|T_{rq}(t) - T_{rq}(t - T_s)\right| \leq 1000\text{Nm/s}^* T_s$

E. Ego Vehicle Dynamic Model

The block takes:

- Torque command
- Initial speed

Outputs:

- Longitudinal speed v_x
- Acceleration a_x(for verification)

IV. IMPLEMENTATION DETAILS

For Q1, the code was designed to investigate how the variation of the TRFC impacts the longitudinal response of the vehicle. The Simulink model remained the same for this question, and the only variation was done with the TRFC value inside the Ego Vehicle block between 1.0 and 0.1. All the parameters of the controllers were kept the same, including the MFC gain η, the ultra-local model gain α, and ADE configuration for capturing the effect of friction. The speed of the vehicle, torque command, and wheel slip ratios were generated as an output in this simulation for every value of TRFC. These have been obtained using To-Workspace blocks and have been plotted outside the model for comparison.

Q2 focuses on the setup and testing of the ADE block. The ADE MATLAB Function utilized a sliding-window buffer whose length N was subject to variation for studying its effect. Three window lengths N = 4, 12, and 24 were tried by simply varying the parameter in the ADE block without any modification to the rest of the model. Furthermore, a purely numerical differentiator has been implemented in parallel to ADE to provide a straightforward comparison of noise-susceptible and algebraic estimation techniques. While running the simulation, both acceleration signals were logged, and the respective plots have been generated.

Q3 was to find the appropriate value of the control gain α for the ultra-local model, which was done analytically, without any model change. Given the provided wheel dynamics and kinematic relationships, α was calculated to be about 0.37. This value was then directly inserted into the MFC MATLAB Function block as the controller's scaling parameter. Substitution was followed by no additional tuning or model modification, so the value of α reflects the physics-based derivation. (Place optional Q3 implementation equation/figure here.) Q4: For this question, the complete MFC-based longitudinal tracking system was assembled and executed with the components that had been setup earlier. The reference velocity profile was fed into the MFC block along with its derivative, and the ADE block supplied the real-time acceleration estimate needed for computing F^(t). The constrained torque command produced by the MFC passed through the torque limiter was then applied to the Ego Vehicle model. In the implementation of this question, no variation of parameters was required, but it validated the integrated working of all subsystems-ADE, MFC, Ego Vehicle, and torque constraints-operating in a closed loop. The logged simulation outputs included the reference and actual velocity trajectories and torque command for visualization. (Place Q4 implementation figure here if included.)

V. SYSTEM ARCHITECTURE

The complete longitudinal velocity control system is implemented in MATLAB/Simulink, following a modular structure aligned with the MFC formulation. The top-level model is shown in Fig. 1, where all major subsystems—MFC controller, ADE estimator, Ego Vehicle dynamics, torque constraint logic, and reference generator—interact in a closed loop.

A. Q1- Effect of Tire–Road Friction Coefficient (TRFC)

Q1 examines how different tire–road friction coefficients influence the vehicle's longitudinal response. Simulations were performed for TRFC = 1.0 and TRFC = 0.1.

The Fig. 4 shows the required torque and velocity and Fig. 5 shows slip ratios for TRFC = 1.0.

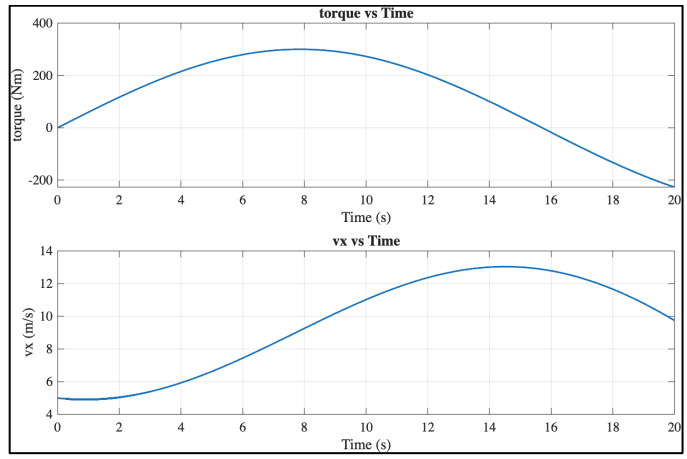

Fig 4 Torque and longitudinal speed (TRFC = 1.0)

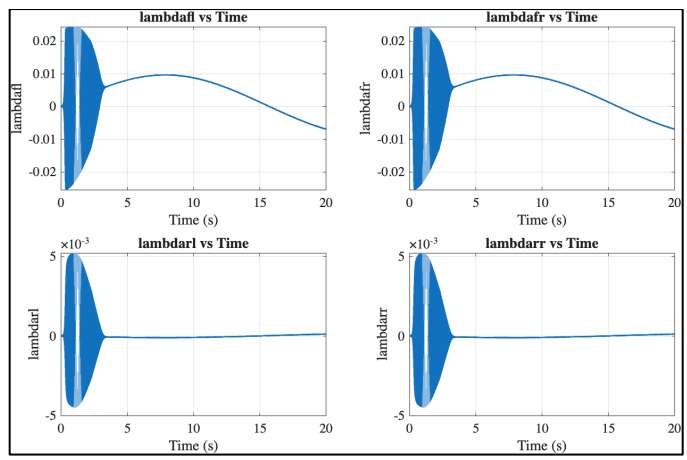

Fig 5 Tire slip ratios (TRFC = 1.0)

The Fig. 6 shows the required torque and velocity and Fig. 7 shows slip ratios for TRFC = 0.1.

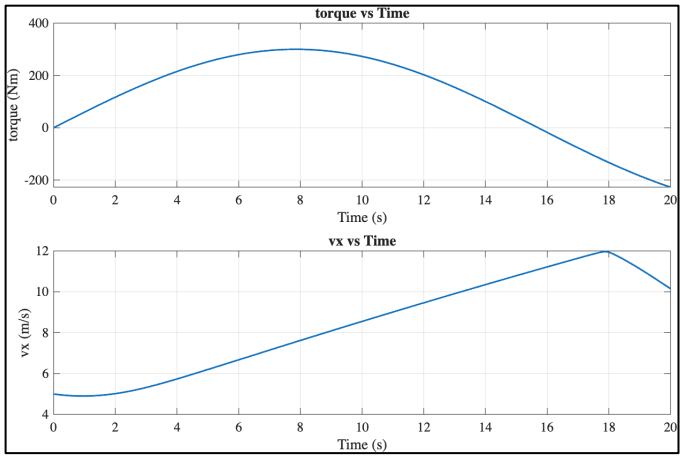

Fig 6 Torque and longitudinal speed (TRFC = 0.1)

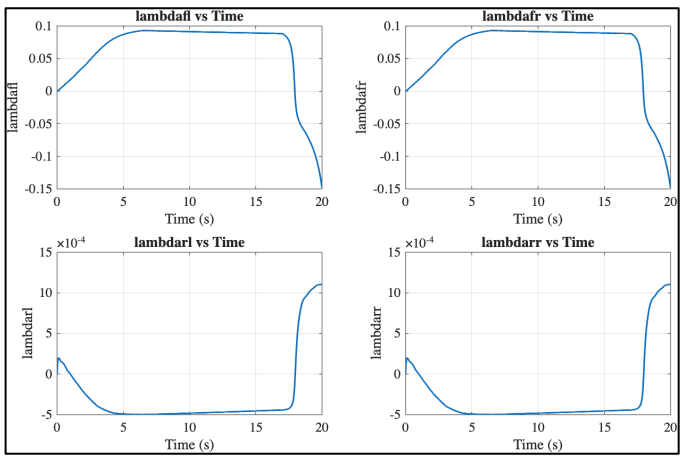

Fig 7 Tire slip ratios (TRFC = 1.0)

B. Q2 - Algebraic Derivative Estimation (ADE) performance

Q2 requires comparing the ADE-estimated acceleration with numerical differentiation and evaluating the effect of the ADE sliding-window size. The result include ADE vs. direct differentiation plot (Fig. 8) and ADE outputs for window sizes N = 4, 12, 24 in Fig. 9, Fig. 10, and Fig. 11 respectively.

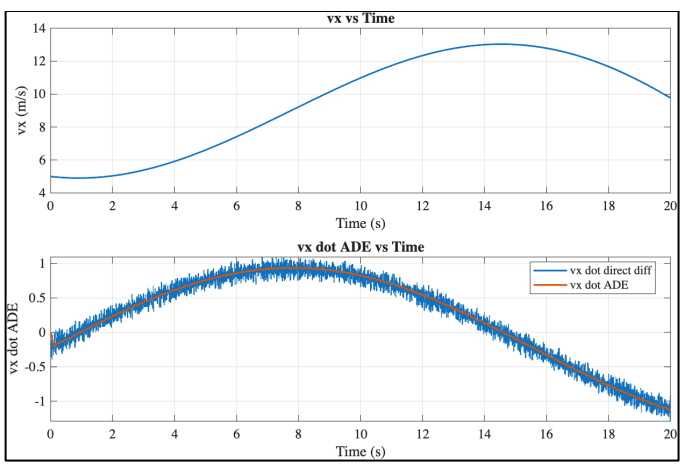

Fig 8 ADE vs. direct differentiation plot

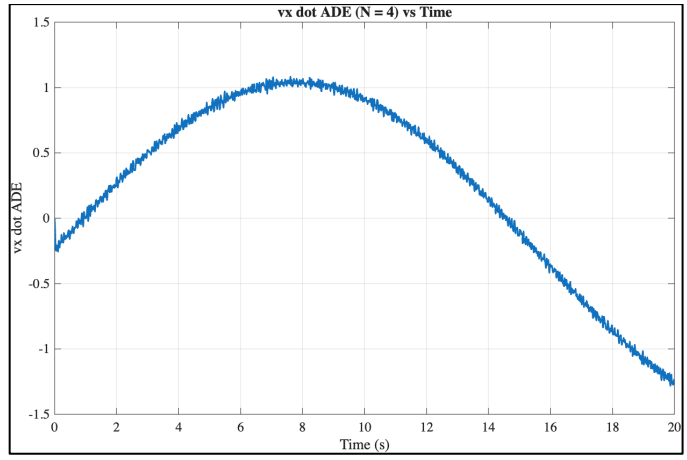

Fig 9 ADE output for window size 4

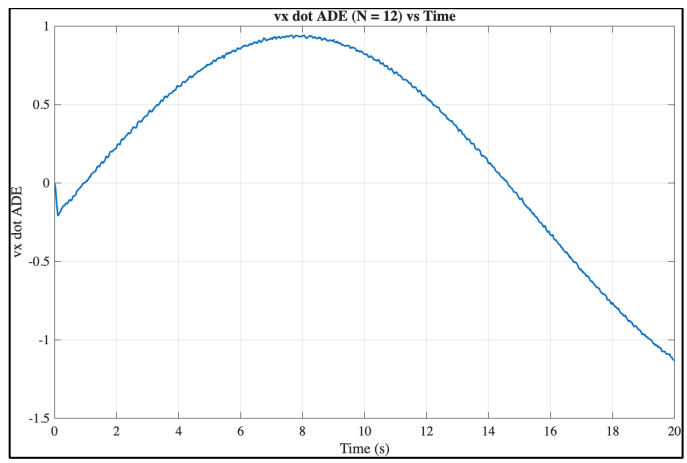

Fig 10 ADE output for window size 12

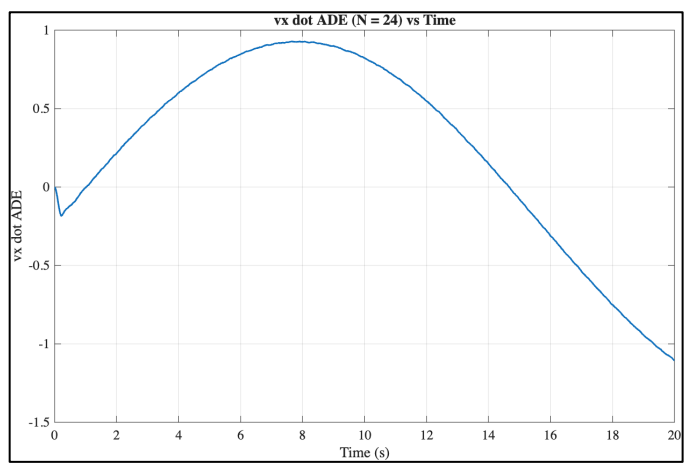

Fig 11 ADE output for window size 24

C. Q3 – Determination of Control Gain α

Q3 requires computing the appropriate control gain α for the ultra-local model used in the MFC controller. Using the wheel rotational dynamics and the kinematic relationship between wheel angular speed and vehicle longitudinal speed, the physical

gain relating applied wheel torque to vehicle acceleration was derived. The final expression

$$\alpha = \frac{R_w}{J_w} \tag{10}$$

was evaluated using the given wheel radius $R_w = 0.33$ m and wheel inertia $J_w = 0.9$ kg\m^2. This yields control gain as 0.37. This value provides the correct physical scaling between torque input and longitudinal acceleration in the ultra-local model.

D. Q4 – Overall Tracking Performance

Q4 evaluates the complete MFC-based longitudinal speed-tracking system using a time-varying reference velocity.

The Fig. 12 shows the required torque input and Fig. 13 shows tracking for actual velocity with the reference velocity for considering the torque constraints.

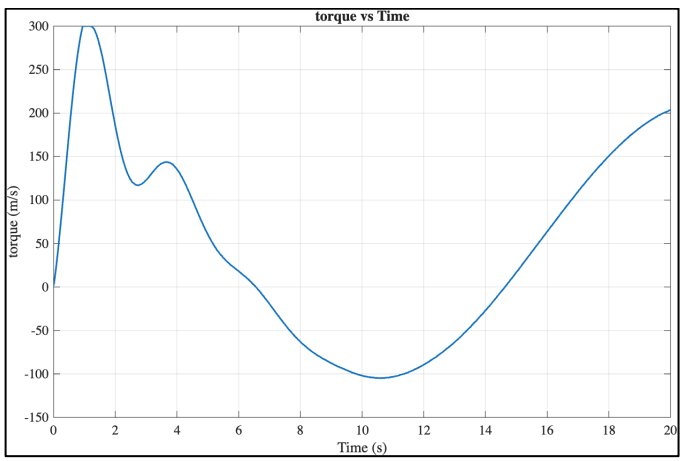

Fig 12 Torque input

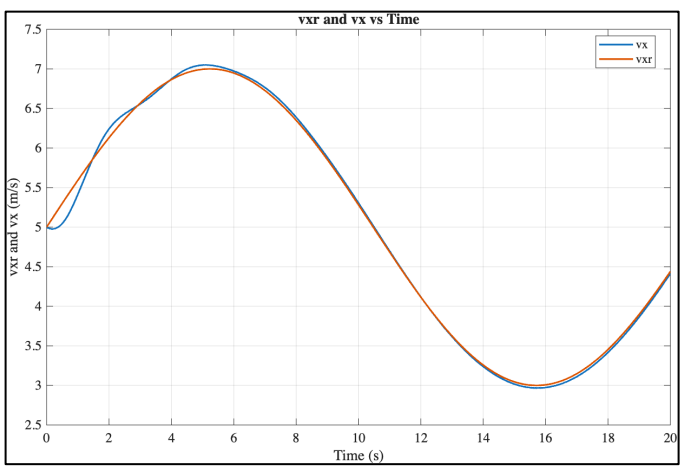

Fig 13 Actual velocity tracking reference velocity

The Fig. 14 shows the required torque input and Fig. 15 shows tracking for actual velocity with the reference velocity without considering the torque constraints.

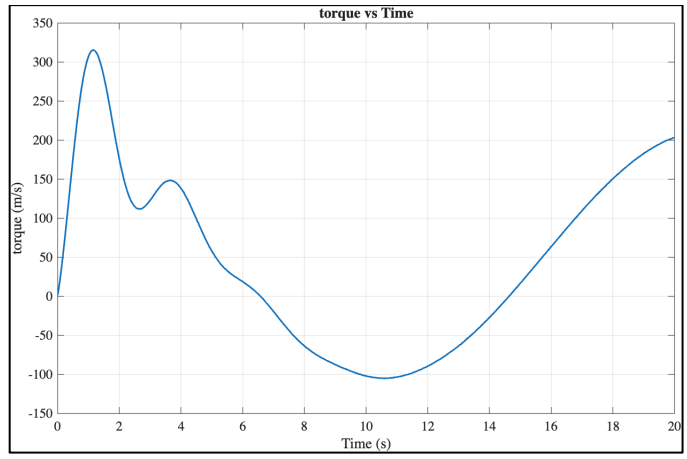

Fig 14 Torque input

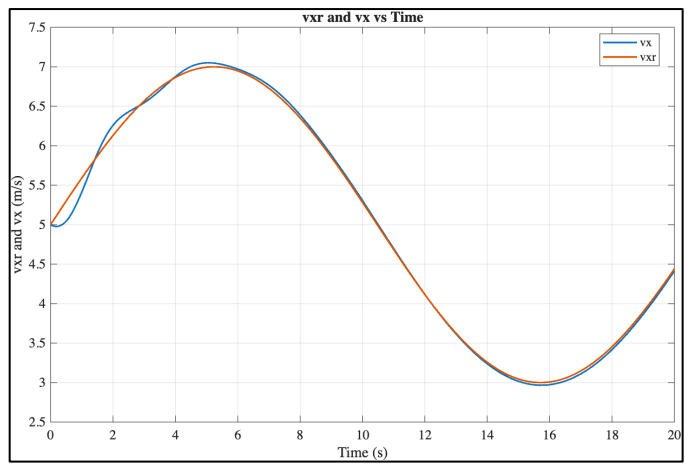

Fig 15 Actual velocity tracking reference velocity

VI. DISCUSSION

The results for Q1 illustrate the behavior that the vehicle assumes for different levels of tire–road friction. For TRFC = 1.0, the applied torque increases smoothly to approximately 300 N·m and decreases in a manner that yields a corresponding increase in vx to almost 14 m/s. All slip ratios remain very small in general (within ±0.02), which gives evidence that the tires operate well within the traction limits. However, an oscillatory pattern for the slip ratios appears in the first few seconds of motion. This can be understood by considering the transitions in the tire dynamics from standstill to rolling motion: the angular velocity of the wheel ramps up faster than the vehicle speed initially, which can create rapid fluctuations in slip until the tire reaches a steady rolling condition. These oscillations diminish rapidly after the vehicle settles into stable acceleration. For TRFC = 0.1, the vehicle exhibits noticeably different behavior. Although the torque input follows a similar shape, the lower friction on the road reduces the maximum achievable longitudinal speed to approximately 12 m/s. The front slip ratios increase drastically, reaching values of approximately 0.1, reflecting partial wheel slip for low friction conditions. By contrast, the rear slip ratios remain much smaller, reflecting the lower demand on the longitudinal load. These results correspond to expected physical behavior: high friction allows smooth

transfer of torque with minimal slip, whereas low friction highly amplifies slip, reducing acceleration capability.

Results for Q2 The performance of the Algebraic Derivative Estimator (ADE) in computing the longitudinal acceleration vx is evaluated for different window lengths N. In the baseline comparison, the ADE estimate agrees nicely with the direct numerical differentiation and manages to smooth the high-frequency noise superimposed on the raw derivative signal without distorting the overall acceleration profile. Employing a small window size, such as N = 4, results in ADE output oscillations with high-frequency noise, which essentially shows poor filtering and poor averaging capability. Increasing the length of the window to N = 12 considerably enhances the smoothness of the estimate, yielding a clean acceleration curve that captures the peak and slope transitions with accuracy. At N = 24, the estimator further smooths and effectively removes most of the noise, though including a slight lag in responding to rapid changes of acceleration. This result further validates the expected trade-off: small N yields responsive but noisy estimates, while larger N produces cleaner but slightly delayed acceleration estimates

Q3 focuses on selecting an appropriate control gain α for the ultra-local model that links wheel torque to vehicle longitudinal acceleration. Using the wheel dynamics and the given physical parameters Rw = 0.33m and Jw = 0.9kg\m, the derived relationship yields $\alpha \approx 0.37$. This value provides a realistic scaling between torque and acceleration, ensuring that the MFC controller neither overreacts nor produces sluggish responses. Although no plots are associated with this question, the computed gain directly influences the stability and responsiveness observed in subsequent simulations. The chosen value offers a balanced and physically consistent behavior for the control algorithm.

Q4 presents a comparison between the performance achieved using the MFC controller with and without torque constraints. In the constrained case, the action of the limiter causes the torque to rise sharply before then moderating its variations. In spite of this, the vehicle speed vx follows the reference vxr in steady-state with little deviation after a short transient has elapsed. If the torque is unconstrained, it evolves with greater freedom but similar magnitude, while the tracking performance shows little difference. In both cases, the vx and vxr curves overlap well. This underlines that the controller provides good reference following without requiring, during this driving cycle, values of torque in excess of those limits imposed.

VII. CONCLUSION

This project aimed to design and validate a complete MFC framework for longitudinal vehicle speed regulation, and the results throughout all questions confirm that this objective was reached. Q1 showed that the vehicle behaves consistently with physical expectations under different tire-road friction levels, which allowed validating the correctness of the overall simulation environment. Q2 confirmed that the Algebraic Derivative Estimator can reconstruct longitudinal acceleration in a reliable way, whose performance can be tuned to balance noise and responsiveness-a necessary requirement for the stability of MFC. Q3 determined a physically meaningful control gain α, which ensures that the ultra-local model links correctly the torque input to vehicle acceleration. Finally, Q4 showed that the implemented MFC controller can track a time-varying speed reference with small steady-state error.

Taking all these results together, it thus follows that the approach used in MFC is not only feasible but also very effective for automotive speed control. It achieves accurate tracking, handles uncertainty, and avoids dependence on detailed vehicle models. Overall, this work validates MFC as a robust model-independent strategy suitable for real-time longitudinal control applications.

APPENDIX

A. *Appendix A – MFC controller*

```
function TORQUE = mfc_torque(v, v_dot_ADE,
alpha, vr, vr_dot, T_last)
% Model-Free Speed Controller (unconstrained)
% Inputs:
%   v          - measured speed
%   v_dot_ADE  - ADE estimate of dv/dt
%   alpha      - ultra-local model gain
(alpha_v)
%   vr         - reference speed
%   vr_dot     - derivative of reference speed

    % tracking error
    e_v = v - vr;

    % ultra-local model: f_hat ≈ v_dot_ADE
    f_hat = v_dot_ADE - alpha*T_last;

    % tuning gain (choose in script or here)
    eta_T = -5;    % you can tune this value

    % unconstrained torque command
    TORQUE = (-f_hat + vr_dot + eta_T * e_v) /
alpha;
end
```

B. *Appendix B – ADE block*

```
function Dot_U = buffer(U, Ts, u_init)
    N =4; % Sliding window length
    persistent y;
    if isempty(y)
        y=u_init*ones(N+1,1); % We initialized
a sliding window here
    end

    y = [U;y(1:end-1)];
    T = Ts*N;
    P = zeros(1,N+1);
    P_alpha = P;
    for k = 1:N+1
        tao_k = (k-1)*Ts;
        P(k) = (6/T^3)*(T - 2*tao_k); % to be
filled
        if((k==1)||(k==N+1))
            P_alpha(k) = P(k)*Ts/2;
        else
            P_alpha(k) = P(k)*Ts;
        end
```

```
    end

    Dot_U =  P_alpha * y; % to be filled

end
```

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